Ela Kumuk

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EDUCATION

Brandeis School of Business and Economics — **Waltham, MA**
Candidate for Master of Science in Business Analytics (STEM-Designated) - GPA: 3.69 — Expected 12/2026
Relevant Coursework: *Machine Learning & Data Analysis for Business and Finance, Generative AI Foundations for Business, Econometrics with R, Python for Business Analytics*

Brandeis University — **Waltham, MA**
BA in Business & BA in Psychology, Minor in Studio Art — 08/2022 - 12/2025

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy), SQL (CTEs, joins, aggregation), R, Advanced Excel (PivotTables, XLOOKUP, Power Query, Macros)
Machine Learning & Generative AI: Machine Learning (Scikit-learn, XGBoost), Feature Engineering, Generative AI, LLM APIs, AI Agents, Prompt Engineering
Analytics: Causal Inference (OLS, Propensity Score Matching, IPW), Walk-Forward Backtesting, Customer Segmentation, Process Automation
Tools: Tableau, Power BI, Bloomberg, Git/GitHub, Jupyter Notebook
Certifications: Databricks Fundamentals (10/2026); Snowflake Data Warehousing Workshop (in progress); Bloomberg Market Concepts (08/2026); IBM Build an AI Agent (07/2026)

WORK EXPERIENCE

MissFlora USA — **Cheyenne, WY**
Business Analyst Intern — 05/2025 - 08/2025
- Evaluated a new US retail channel from market share and pricing data (SQL); ownership adopted my recommendation
- Analyzed sales across 20 CPG SKUs in Python (Pandas), identifying the top revenue drivers to guide product placement
- Automated a manual supply-chain reporting process in Python, saving ~4 hours of weekly work

Kumukh Parfums — **Istanbul, TR**
Founder — 09/2021 - 09/2024
- Founded and ran a niche perfume brand: developed 2 fragrance lines at a $100 price point and sold 200+ units ($20K+ revenue), owning sourcing, packaging, pricing, and every customer relationship
- Turned a defective 30-unit batch into a retention channel; past customers who received it bought ~2.5 units each afterward

PROJECTS

State Street Investment Management — **Boston, MA**
Graduate Field Project (Brandeis, 4-person team): Multi-Asset Meta-Labeling Pipeline — 05/2026 - 08/2026
- Ran a factor-weighting sweep on the team's multi-asset allocation model: a risk-heavy weighting lifted walk-forward Sharpe from 0.71 to 0.81, positive in all 6 out-of-sample folds
- Flagged that the secondary ML sizing model was near chance out of sample (AUC ~0.49) rather than reporting only wins
- Presented findings with the team to a State Street stakeholder and our faculty advisor

Causal Inference: LLM Reasoning vs. Econometric Specification — **Waltham, MA**
Marketing Analytics Course Project — Spring 2026
- Estimated a non-randomized campaign's effect on 5,000 users in Python using OLS, propensity score matching, and IPW; correctly specified models converged on ~+5.5 pp
- Showed that controlling for post-treatment mediators (impressions, clicks) collapsed the estimate to ~0, and benchmarked two LLMs that failed in opposite ways

Ovrule - AI Agent Governance Platform | decision-receipt-lab.vercel.app — **Cambridge, MA (Remote)**
Creator & Sole Developer — 04/2026 - Present
- Built an AI-agent audit tool (Next.js, Supabase) whose LLM classifier checks agent decisions against six rules
- Released as an open-source npm package (ovrule-lab) with a public demo; Top 7 finalist, Ship Season S26 buildathon

PUBLICATIONS, ACTIVITIES & SKILLS

Publication: Co-author, "Use of Velocity Vectors for Cell Classification Under Acoustic Drifting Forces," IEEE TIPTEKNO, 2020
Activities: National Association for Business Economics - NABE (2026 - Present), Brandeis Women in Economics & Finance (2026 - Present), Brandeis Entrepreneurship & Technology Assoc. (2022 - 2024). **Volunteer:** Certified Ballet Instructor - Teaching ballet to kids with no prior access to the arts in Turkey (2021 - 2022).
Languages: Turkish (Native), English (Fluent), French (Beginner)